bayesAgriFlow Experimental Design Cheatsheet

1/10 — Central Composite Design (CCD): Structure and Purpose

Bayesian response-surface workflows for agronomic experiments

bayesAgriFlow

1 What is a CCD?

A Central Composite Design (CCD) is used to fit **second-order response surfaces** efficiently when curvature is expected.

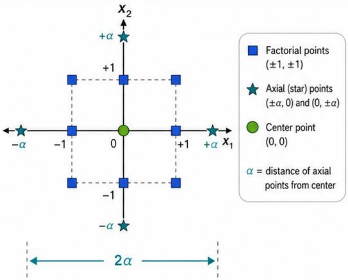
$$\text{Runs} \approx 2^k + 2k + n_0$$

where: k = number of factors,
n_0 = number of center-point replicates.

2 Core building blocks

- Factorial points
- Axial (star) points
- Center points
- Coded factors (−1, 0, +1, ±alpha)

3 2-factor geometry



4 Why use CCD?

- Estimates linear, interaction, and quadratic terms
- Detects curvature
- Supports optimization
- Flexible for sequential experimentation

5 When to use it

Need a response surface?	yes
Expect curvature?	yes
Need an optimum?	yes
Limited runs?	CCD is often a strong choice

6 Key functions

```
bayes_ccd()
bayes_rsm_audit()
bayes_rsm()
bayes_rsm_optimum()
```

7 Typical workflow

Choose factors → Generate CCD → Collect data → Fit response surface → Interpret curvature → Optimize

Key takeaway

CCD is the foundational design for Bayesian response-surface analysis in **bayesAgriFlow** because it balances information, flexibility, and practical run size.

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10/10 — Integrated Workflow for Advanced Experimental Design

How the functions connect from planning to decision

bayesAgriFlow

1 Design selection map

Need quadratic surface? → Yes → CCD or Box-Behnken

No ↓

Need criterion-based design? → Yes → Optimal design

No ↓

Need flexible curvature? → Yes → Pseudoquadratic design

2 End-to-end workflow

Select factors → Generate candidate designs → Audit efficiency → Run experiment

Fit response surface → Optimize → Augment if needed

3 Function map

Design generation	Diagnostics	Modeling	Decision
bayes_ccd()	bayes_rsm_audit()	bayes_rsm()	bayes_rsm_optimum()
bayes_box_behnken()	bayes_design_efficiency()	bayes_pseudoquadratic()	bayes_rsm_economic_optimum()
bayes_optimal_design()	bayes_design_rank()	bayes_rsm_canonical()	bayes_rsm_desirability()
			bayes_rsm_augment()

4 Bridge to external workflows

Import external designs, results, or data structures with `bayes_rsm_import()`.

Compare response-surface workflows and outcomes using `bayes_compare_rsmflow()`.

5 Best practices

- **Code factors carefully:** Use sensible ranges and coding (−1, 0, +1) for interpretability and numerical stability.
- **Inspect curvature early:** Audit designs and visualize to confirm the model can capture curvature adequately.
- **Quantify uncertainty:** Use diagnostics and replication to assess precision and robustness.
- **Document design choices:** Record rationale, settings, and results for transparency and reproducibility.

Key takeaway

The advanced experimental-design module works best as a **connected workflow** rather than isolated functions.

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3/10 — Fitting Bayesian Response Surfaces from CCD

From design matrix to posterior response surface

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1 Second-order model

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \beta_{ii} x_i^2 + \sum_{i<j}^{k} \beta_{ij} x_i x_j + \varepsilon$$

y = response

x_i = coded factor i (mean 0, typically in $[-\alpha, +\alpha]$)

β terms = regression coefficients

$\varepsilon \sim Normal(0, \sigma^2)$ = random error

2 What the terms mean

Linear effects ($\beta_i x_i$)
Change in response for a one-unit change in factor i.

Interaction effects ($\beta_{ij} x_i x_j$)
How the effect of factor i depends on factor j (and vice versa).

Quadratic effects ($\beta_{ii} x_i^2$)
Curvature—response increases then decreases (or vice versa).

3 Workflow

4 Posterior outputs

Posterior means
Estimated coefficients ($E[\beta \mid \text{data}]$).

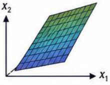
Credible intervals
Uncertainty for coefficients and predictions.

Fitted response surface
The posterior mean surface over factor space.

Predicted responses
Posterior predictive distribution at new factor settings.

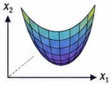
5 Reading the surface

Uphill slope (linear effect)



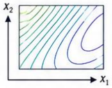
Response increases as you move along the factor.

Curvature (quadratic effect)



Response bends—there is an optimum or turning point.

Interaction (effect depends on another factor)



Contours tilted or warped indicate interaction.

6 Key functions

```
bayes_rsm()
bayes_pseudoquadratic()
bayes_predict()
```

7 Frequentist vs Bayesian view

Frequentist view	Bayesian view
Point estimate (single best value)	Posterior distribution (complete uncertainty)
$p = 0.03$ — p-value (dichotomous decision)	Credible interval & probability statements (e.g., $P(\beta_i > 0 \mid \text{data}) = 0.93$)

Key takeaway

CCD becomes most powerful when the fitted second-order model is interpreted as a full response surface rather than a set of isolated coefficients.

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7/10 — Optimal Designs: A-, D-, I-, and G-optimality

Designing experiments for information efficiency

bayesAgriFlow

1 Why optimal designs?

Optimal designs place experimental runs where they are most informative, yielding more precise estimates or predictions for the same budget.

- Maximize information per run
- Improve precision or robustness
- Handle complex models efficiently
- Support defensible experimental choices

2 Four core criteria

- **A-optimality:** minimize average parameter variance
- **D-optimality:** maximize determinant-based information
- **I-optimality:** improve average prediction precision
- **G-optimality:** control worst-case prediction variance

3 How to choose a criterion

Start from your goal.

- **Estimation** (parameters) → A-optimality
- **Prediction** (average case) → I-optimality
- **Protection** (weak regions) → G-optimality

When in doubt, compare multiple criteria and select the best trade-off.

4 Mini comparison table

Criterion	Emphasizes	Strengths	Trade-offs
A	Average parameter variance	Precise parameter estimates	May ignore poor prediction in extreme regions
D	Overall information volume	Good general efficiency; robust in many cases	Can still allow high variance in some regions
I	Average prediction variance	Directly targets prediction precision	Less focused on parameter estimation
G	Worst-case prediction variance	Guards against poorly informed regions	Often more conservative; may use runs less efficiently

5 Key functions

```
bayes_optimal_design()
bayes_compare_designs()
bayes_design_information()
```

Find optimal designs, compare candidates, and quantify information for your model.

6 Agronomic use cases

Fertilizer trials
- Efficiently estimate response to nutrient rates and interactions

Multi-factor greenhouse studies
- Optimize limited runs across temperature, light, water, and genotype

Costly field experiments
- Safeguard against weak regions and maximize learning per plot

Key takeaway | Optimal design criteria let you tailor the experiment to the scientific question instead of relying on a single default layout.

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5/10 – CCD Optimization, Assurance, and Augmentation

Turning a fitted surface into biological and economic decisions

bayesAgriFlow

1 Optimization targets

Biological optimum
Maximize the mean response (e.g., yield, biomass).

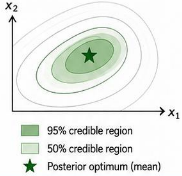
Economic optimum
Maximize expected profit (revenue – cost).

Desirability optimum
Balance multiple objectives and constraints.

2 Posterior optimum

The optimum is a random point due to parameter uncertainty.

Reported as a posterior distribution (e.g., mean, median, credible region).



3 Assurance

Assurance quantifies how well the planned design will perform.

- Expected precision at the optimum (e.g., expected variance).
- Confidence that the true optimum meets a target performance.
- Used to compare and validate candidate designs.

4 When to augment

- Poor precision at the optimum (high expected variance).
- Optimum lies near or outside the design region.
- Wide credible region.
- Unresolved curvature (model uncertainty remains high).

5 Sequential strategy

Initial CCD (explore space) → Fit surface (estimate model) → Inspect uncertainty (assess optimum) → Add runs (augment design) → Refit (update model)

6 Key functions

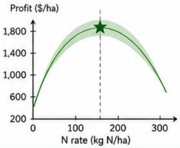
```
bayes_rsm_economic_optimum()
bayes_rsm_desirability()
bayes_rsm_assurance()
bayes_rsm_augment()
```

7 Agronomic example

Response:	Corn grain yield (t/ha)
Factor:	Nitrogen rate (kg N/ha)
Target:	Maximize profit
Fertilizer cost:	$1.20 per kg N
Corn price:	$180 per ton

Result (economic optimum):

N* = 168 kg N/ha
Expected profit = $1,248/ha
95% credible region: 150–186 kg N/ha



Key takeaway

Optimization is not only about finding the best point;
it is also about quantifying how certain that recommendation is.

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9/10 — Pseudoquadratic Models for Agronomic Response Surfaces

Flexible alternatives when the usual quadratic form is too rigid

bayesAgriFlow

1 Why pseudoquadratic?

Agronomic responses may be **asymmetric** or change scale **nonlinearly** with factor levels.

Standard quadratics assume symmetry and constant curvature, which can be too restrictive.

2 Core idea

Replace squared terms with **transformed powers** of the factors.

These powers control how quickly the response bends, allowing asymmetric and scale-flexible curvature.

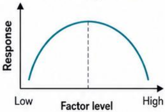
Example: Use $x^{p_1}, x^{p_2}, x_1^{p_1} x_2^{p_2}$ with $0 < p_i \le 2$
($p_i = 2$ reduces to the standard quadratic)

3 When to consider it

- **Skewed response shape**
  The peak or valley is off-center.
- **Unequal curvature**
  Curvature differs along axes or directions.
- **Dose–response behavior**
  Diminishing or accelerating returns.
- **Fertilizer studies**
  Biological responses often show asymmetry and thresholds.

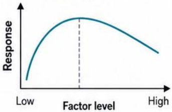
4 Visual comparison

Standard quadratic



Symmetric peak, equal curvature on both sides.

Pseudoquadratic



Asymmetric shape, unequal curvature.

5 Modeling steps

Choose design → Fit quadratic → Inspect mismatch → Fit pseudoquadratic → Compare surfaces

If mismatch remains, try different power settings p_i or reparameterize factors.

6 Key functions

```
bayes_pseudoquadratic()
bayes_rsm()
bayes_compare_designs()
```

7 Interpretation note

Focus on how the surface behaves (shape, slope, and turning) and what it means agronomically. Avoid over-interpreting individual coefficients.

8 Typical power guidelines

$p_i = 2$	Standard quadratic (symmetric)
$p_i \approx 1.5$	Moderate asymmetry
$p_i = 1$	Linear-like (diminishing curvature)
$p_i < 1$	Strong early change, flattening later

Key takeaway

Pseudoquadratic models extend response-surface analysis when biological behavior is more flexible than a symmetric quadratic.

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6/10 — Box-Behnken Designs

An efficient alternative for second-order response surfaces

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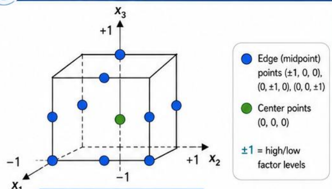
1 What is a Box-Behnken design?

A Box-Behnken design (BBD) is a response surface design that places experimental runs at the **midpoints of edges and at the center** of the factor space, avoiding extreme corner points.

2 Main characteristics

- No extreme corner points (safer operating region)
- Efficient for fitting quadratic models (main effects + interactions + curvature)
- Balanced, rotatable or nearly rotatable structure with good prediction properties
- Includes center points for pure error and lack-of-fit assessment

3 3-factor geometry



Total runs for k factors:

$$N = 2k(k-1) + n_0$$

where n_0 = number of center replicates

4 When is it preferable to CCD?

- When experiments have safety or operational limits
- When extreme factor combinations are undesirable or infeasible
- When a quadratic model is needed with moderate run size
- When resources are limited but curvature must be detected

5 Strengths and limitations

Strengths

- Avoids extreme (corner) points
- Efficient for quadratic model estimation
- Good prediction within the design region
- Robust to model misspecification
- Rotatable (or near-rotatable) versions available

Limitations

- Does not explore corners (limits extrapolation towards extremes)
- Slightly less efficient for pure curvature than CCD in some cases
- Less suited when extreme settings are of primary interest

6 Key functions

```
bayes_box_behnken()
bayes_rsm()
bayes_compare_designs()
```

7 Choose CCD or BBD?

Consideration	Choose BBD	Choose CCD
Extreme settings allowed?	No / Limited	Yes
Need to explore corners?	No	Yes
Primary goal	Fit quadratic model safely	Explore a wide region
Typical run size (3 factors)	15–21 runs	20–30 runs

Key takeaway

Box-Behnken designs are useful when a quadratic model is needed but extreme factor combinations are undesirable.

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4/10 — Canonical Analysis and Stationary Points

Interpreting the geometry of a fitted CCD surface

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1 What is a stationary point?

A stationary point is a point on the fitted response surface where the **slope is zero** in every direction:

$$\nabla \hat{y}(\boldsymbol{x}^*) = 0$$

It can be a maximum, minimum, saddle, or (rarely) flat region.

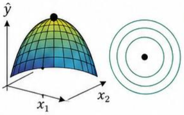
2 Canonical analysis

1. **Shift origin:** translate coordinates so the stationary point is at $(0, 0)$.
2. **Inspect eigenvalues:** compute the eigenvalues of the Hessian matrix H.
3. **Classify the surface:** use eigenvalue signs to determine surface type.

3 Surface types

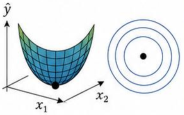
Maximum

Curves downward in all directions.



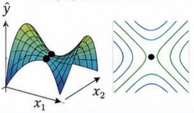
Minimum

Curves upward in all directions.



Saddle

Curves upward in one direction and downward in the perpendicular direction.



4 How to interpret eigenvalues

λ

All negative $(\lambda_1 < 0, \lambda_2 < 0)$	**Maximum**
All positive $(\lambda_1 > 0, \lambda_2 > 0)$	**Minimum**
Mixed signs $(\lambda_1 \times \lambda_2 < 0)$	**Saddle**

If any eigenvalue is ~ 0, the surface is flat or nearly flat in that direction (investigate ridge behavior).

5 Practical interpretation

Optimum dose
Minimum surface → an optimal factor combination likely exists.

Non-optimal ridge
Flat or mixed curvature → many combinations give similar responses.

Unstable response region
Saddle surface → small changes can increase or decrease response.

6 Key functions

```
bayes_rsm_canoncal()
bayes_rsm_optimum()
bayes_rsm_steepest()
bayes_rsm_ridge()
```

7 Decision pathway

Fit model (CCD) → Locate stationary point → Classify surface (eigenvalues) → Decide whether to optimize or augment

Key takeaway

Canonical analysis tells you whether the fitted response surface really contains a useful optimum or only a saddle-shaped region.

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8/10 — Measuring Design Efficiency and Prediction Quality

From matrix diagnostics to practical design ranking

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1 What should be checked?

- **Rank** — Sufficient information (full column rank)
- **Estimability** — All target functions estimable
- **Orthogonality** — Low correlation among factor estimates
- **Prediction variance** — Small and uniform across region
- **Cost** — Information gained per unit cost

2 Efficiency metrics

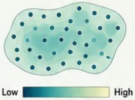
	Metric	Formula	Meaning
A	A-efficiency	$\phi_A = \frac{\mathrm{tr}(M^{-1})_{\mathrm{ref}}}{\mathrm{tr}(M^{-1})}$	Smaller average variance
D	D-efficiency	$\phi_D = \left(\frac{\lvert M\rvert}{\lvert M\rvert_{\mathrm{ref}}}\right)^{1/p}$	Smaller generalized variance
I	I-efficiency	$\phi_I = \frac{\mathrm{tr}(M_{\mathrm{xef}}^{-1} M)}{p}$	Average information relative to reference
G	G-efficiency	$\phi_G = \frac{1}{\max_{x \in \mathcal{R}} g(x, M)}$	Controls worst-case variance
R	Relative efficiency	$RE = \frac{\phi_{\mathrm{design}}}{\phi_{\mathrm{ref}}}$	Compare designs or benchmarks

$M = X^{\top}X$ (information matrix), p = number of estimated parameters

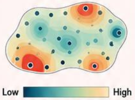
3 Prediction quality

Prediction variance across the design region $\mathcal{R}$

Low prediction variance (preferred)



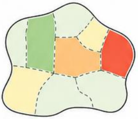
High prediction variance (not preferred)



4 Region sensitivity

A design can look efficient on average but still have large prediction variance in certain subregions.

- Check maps of prediction variance.
- Identify "hotspots" of high uncertainty.
- Add runs strategically to improve weak areas.



- Low variance
- Moderate
- High variance

5 Cost-efficiency

Good designs balance information gain and experimental cost.

$$\text{Cost-efficiency} = \frac{\text{Information gain}}{\text{Total cost}}$$

- Higher is better.
- Use costs per run, factor settings, or scenario evaluations.

6 Key functions

Function	Purpose
`bayes_design_efficiency()`	Compute efficiency metrics for a design
`bayes_design_rank()`	Rank multiple designs by chosen metrics
`bayes_design_region_sensitivity()`	Assess prediction variance by subregions
`bayes_design_cost_efficiency()`	Evaluate information gain per cost

7 Practical workflow

Generate candidates — Create a set of feasible designs → **Audit** — Check rank, estimability, orthogonality → **Rank** — Compute metrics and rank designs → **Inspect region sensitivity** — Map prediction variance and find weak areas → **Choose final design** — Select the best trade-off of efficiency and cost

Key takeaway | A good design is not just estimable; it should also be efficient, stable across the region, and worth its cost.

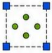
bayesAgriFlow Experimental Design Cheatsheet

2/10 – CCD Variants, Coding, and Run Planning

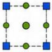
Choosing alpha, coding factors, and planning practical experiments

bayesAgriFlow

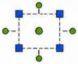
1 CCD variants



Circumscribed (CCC)
Star points lie inside the factorial cube.
Smaller experimental region.



Face-centered (CCF)
Star points lie on the faces.
Common and easy to implement.



Inscribed (CCI)
Star points lie outside the factorial cube.
Requires wider factor ranges.

2 Coding real factors

$$x = \frac{X - \text{center}}{\text{step}}$$

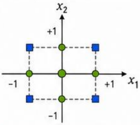
Example (Nitrogen dose)

Center (optimum guess)	X = 120 kg N/ha
Step (change for 1 coded unit)	step = 30 kg N/ha
Real value	X = 150 kg N/ha
Coded value	x = (150 – 120)/30 = 1.0

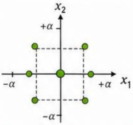
Use the same coding rule for all factors.

3 How alpha changes the geometry

alpha = 1 (face-centered)



alpha > 1 (inscribed)



Larger α ⟶ star points farther from center ⟶ more curvature sensitivity

4 Planning the run size

Factorial points	2^k	All combinations at coded levels (–1, +1)
Axial points	$2k$	One at $+\alpha$ and one at $-\alpha$ for each factor
Center replicates	n_0	At (0, 0, ..., 0) to estimate pure error

Total runs $N = 2^k + 2k + n_0$

Choose $n_0 \geq 3$ for better variance estimation.

5 Practical agronomic questions

- **Factor ranges:** What are realistic minimum and maximum levels?
- **Safe operating region:** Are star points within safe and feasible limits?
- **Curvature:** Do we expect nonlinearity in the response?
- **Replication:** How many center replicates are needed for precision?

6 Key functions

```
bayes_ccd()
bayes_rsm_audit()
bayes_rsm_import()
```

7 Design tips

- *Randomize run order to reduce bias from time or field effects.*
- *Replicate center points to estimate pure error and check stability.*
- *Keep star points within feasible and agronomically safe ranges.*
- *Start with alpha = 1 (CCF); increase alpha only if more curvature exploration is needed.*

8 Variant quick guide

Variant	Star location	Pros	Consider when...
CCC		Compact region, good for safety	Factor ranges are narrow or risky
CCF		Simple, widely used	Ranges are moderate
CCI		Explores extremes, more curvature info	Wide ranges are feasible

Key takeaway

Choose the **CCD variant** that matches the experimental region, coding scheme, and practical constraints of the agronomic study.

bayesAgriFlow Spatiotemporal Models Cheatsheet

1/5 — Foundations and Unified Workflow

Bayesian space–time modeling for repeated agricultural observations

bayesAgriFlow

1 Why spatiotemporal?

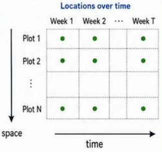
- Repeated observations across field locations and dates.
- Spatial dependence among nearby plots.
- Temporal dependence across repeated measurements.
- Improved prediction, monitoring, and decision support.

2 Core design object

`bayes_spatiotemporal()`

Required roles:

- ✓ `response` → response variable
- ✓ `x_coord` → x coordinate (easting)
- ✓ `y_coord` → y coordinate (northing)
- ✓ `time` → time (e.g., week or date)
- ✓ `group/block` → optional grouping (e.g., block, cultivar, management)



3 Model components

response (*y*) = mean structure (*fixed effects*) + spatial GP (*space*) + temporal dependence (*time*) + residual error (*noise*)

Fixed effects (intercept, covariates)

Spatial Gaussian Process (captures correlation across locations)

Temporal structure (AR(1), factor-time, linear-time, or no-time)

Residual error (independent noise)

Temporal dependence options (sensitivity): AR(1) | Factor-time (unstructured by time level) | Linear-time (trend over time) | No-time (independent across times)

4 Stable workflow

Prepare data — Clean, organize, and check data quality.
→ **Define design** — `bayes_spatiotemporal()` Specify roles and model structure.
→ **Fit model** — `bayes_fit()` Estimate posterior distributions.
→ **Diagnose** — `bayes_diagnose()` Check convergence, fit, and assumptions.
→ **Predict / Forecast** — `bayes_st_predict()` `bayes_st_forecast()` Predict now or ahead.
→ **Decide** — Interpret results and take action.

5 When to use it

- *Disease progression in trial plots.*
- *Soil moisture monitoring.*
- *Repeated canopy temperature.*
- *Seasonal yield-risk mapping.*

6 Key functions

- `bayes_spatiotemporal()`
- `bayes_st_monitor()`
- `bayes_st_covariance()`
- `bayes_st_parameters()`
- `bayes_st_predict()`
- `bayes_st_forecast()`

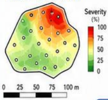
7 Agronomic mini-example

Scenario: Wheat disease severity measured weekly in 24 plots.
Inputs: x-y location, week, cultivar.
Outputs: Posterior field map and 1-week-ahead forecast.

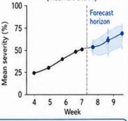
Example data (first 6 rows)

plot_id	x (m)	y (m)	week	cultivar	severity
1	12.4	8.7	1	A	3.1
2	24.6	7.3	1	A	4.0
3	36.8	6.9	1	B	2.8
4	48.6	9.2	1	B	3.6
5	60.7	8.1	1	A	4.2
6	72.9	7.6	1	B	2.6
...	...	...	...	...	...

Posterior field map (Week 8)



1-week-ahead forecast (mean ± 90% PI)



Goal: estimate current risk and predict short-term spread.

Key takeaway

Spatiotemporal models are most useful when agricultural observations are structured by both place and time.

bayesAgriFlow Spatiotemporal Models Cheatsheet

5/5 — Change Detection and Agronomic Decisions

Using posterior change and risk products for management decisions

bayesAgriFlow

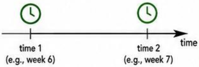
1 Why change matters

- Crop stress evolves over time.
- Interventions must be timely.
- Posterior change quantifies evidence.
- Management can be spatially targeted.

2 Matched-location change

```
bayes_st_change()
```

Compares the same locations at two times and estimates the posterior probability of increase or decrease.



Output: posterior probability of increase, decrease, or stability at each location.

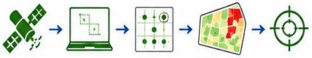
3 Reading a change map

- Red: strong evidence of decrease.
- Yellow: little change or uncertain.
- Green: strong evidence of increase.

4 Decision rules

Situation	Posterior evidence	Possible action
Disease increase	P(increase) ≥ 0.75	Apply or schedule treatment
Moisture decline	P(decrease) ≥ 0.75	Irrigate or adjust management
Stable conditions	P(stable) ≥ 0.70	Continue monitoring; no action

5 From model to action



Monitor → **Fit** → **Predict** → **Detect change** → **Prioritize intervention**

- **Monitor**: Collect repeated field observations
- **Fit**: Fit spatiotemporal model to data
- **Predict**: Generate posterior maps for times of interest
- **Detect change**: Compute posterior change maps
- **Prioritize intervention**: Focus resources where change risk is highest

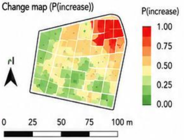
6 Key functions

`bayes_st_change()`	Compute posterior change (increase/decrease/stable) between two times.
`bayes_st_predict()`	Predict posterior distribution of the response at future times.
`bayes_st_pxceedance()`	Posterior probability that response exceeds a threshold at each location.
`bayes_decide()`	Convert evidence and rules into recommendations or actions.

7 Agronomic case vignette

Scenario: fungicide targeting in a wheat field

Compare week 6 vs week 7.



Interpretation

High-probability increase (P ≥ 0.75) detected in the northeast sector.

Recommendation

Prioritize scouting and consider targeted fungicide application in the northeast zone.

Key takeaway

Posterior change analysis helps convert spatiotemporal evidence into timely agronomic action.

bayesAgriFlow Spatiotemporal Models Cheatsheet

3/5 — Monitoring, Diagnostics, and Parameter Interpretation

Checking data coverage and understanding posterior components

bayesAgriFlow

1 Before fitting

`bayes_st_monitor()`

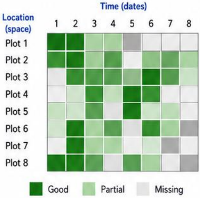
Check for missing dates
Identify dates with no observations.

Check for sparse locations
Find plots or areas with very few observations.

Check for unbalanced coverage
Ensure even sampling across space and time.

Check for duplicated coordinates
Detect repeated or near-identical location pairs.

2 Coverage audit



Good coverage looks like

- Most plots sampled on most dates.
- Few gaps and no long missing stretches.
- Balanced coverage across space and time.

Warning signs

- Entire dates with many missing plots.
- Many plots missing for several dates.
- Gaps concentrated in specific areas or time periods.

3 After fitting

`bayes_diagnose()`

Check convergence
Inspect trace plots and R-hat.

Check effective sample size
Ensure ESS is sufficiently large.

Posterior predictive checks
Compare replicated vs. observed data.

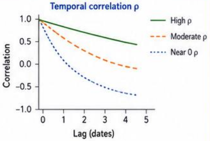
Inspect residual patterns
Look for spatial or temporal structure in residuals.

4 Parameter summaries

`bayes_st_parameters()`

Parameter	Symbol	Meaning	Plain-language interpretation
Fixed effects	β	Mean structure for covariates	Average effect of predictors on the response.
Spatial range	ϕ	How quickly locations become less similar	Larger ϕ = broader spatial influence (longer range).
Temporal correlation	ρ (rho)	Autocorrelation across consecutive dates	Higher ρ = stronger persistence over time.
Residual SD	σ	Unstructured noise (micro-scale variation)	Typical unexplained variation left after accounting for structure.

5 What does rho mean?



- **ρ near 0:** weak serial dependence — values change quickly.
- **ρ moderate:** some persistence — today informs tomorrow.
- **ρ high:** strong memory — values remain similar across dates.

6 Key functions

`bayes_st_monitor()`

`bayes_fit()`

`bayes_diagnose()`

`bayes_st_parameters()`

`bayes_summary()`

Use these functions in sequence to monitor data, fit the model, diagnose results, and interpret parameters.

7 Teaching example

Scenario: soil moisture across a field over 8 sampling dates.
Model: Gaussian process, AR(1) in time.

Posterior summary (median [95% CrI])

Parameter	Symbol	Median	95% CrI
Intercept	β_0	18.6	[15.4, 21.9]
Soil texture (clay %)	β_1	0.42	[0.22, 0.63]
Spatial range	ϕ (km)	1.85	[1.10, 3.10]
Temporal correlation	ρ	0.64	[0.42, 0.82]
Residual SD	σ	2.35	[1.90, 2.95]

Findings:

- Strong spatial structure ($\phi \approx 1.9$ km).
- Moderate temporal persistence ($\rho \approx 0.64$).
- Diagnostics show good convergence and PPC support.

Key takeaway

Good diagnostics turn a fitted space–time model into a trustworthy scientific result.

bayesAgriFlow Spatiotemporal Models Cheatsheet

2/5 — Covariance Structures and Temporal Options

Choosing the right space–time structure for agricultural monitoring

bayesAgriFlow

① Why covariance matters

- Nearby plots are more similar.
- Repeated dates are correlated.
- Covariance structure controls smoothing.
- Misspecification affects prediction and uncertainty.

② Separable structure

`bayes_st_covariance()`

Core idea:

$$\mathrm{Cov}(\text{space, time}) = K_{space} \otimes K_{time}$$

- ✓ **Efficient:** reduces parameters and computation.
- ✓ **Interpretable:** separates spatial and temporal processes.
- ✓ **Easy to compare:** swap temporal options while keeping spatial part fixed.

③ Spatial component

- **Similarity decreases with distance.**
- **Gaussian process intuition:** nearby locations share information.
- **Range controls** how far correlation extends, **sill** sets overall variation, **nugget** captures fine-scale noise.

④ Temporal options

Option	Best for	Strength	Limitation
`AR(1)`	Dense, regularly spaced time series.	Captures persistence with few parameters.	Assumes constant correlation between adjacent times.
`factor-time`	Irregular or seasonal patterns.	Very flexible; no ordering assumption.	More parameters with many time levels.
`linear-time`	Clear increasing or decreasing trend.	Captures trend parsimoniously; easy to interpret slope.	May miss non-linear or periodic behavior.
`no-time`	Weak or absent temporal signal.	Simplest; fewer parameters.	Ignores temporal correlation; may underfit if signal exists.

⑤ How to choose

- Dense repeated dates (e.g., weekly)? → Use `AR(1)`
- Irregular time patterns or seasons? → Try `factor-time`
- Monotone trend over time? → Try `linear-time`
- Weak temporal signal or very noisy? → Compare against `no-time`

⑥ Key functions

`bayes_st_covariance()`
Specify spatial and temporal covariance.

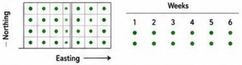
`bayes_spatiotemporal()`
Build the spatiotemporal model.

`bayes_fit()`
Fit the model and obtain posteriors.

`bayes_compare_models()`
Compare structures (WAIC, PSIS-LOO, etc.).

⑦ Agronomic mini-example

Scenario: Canopy temperature (°C) across 40 georeferenced plots over 6 weeks.



Candidate structure	Parameters (approx.)	PSIS-LOO (elpd)	Δ elpd vs best
AR(1) + spatial GP	29	−1243.6	0.0
Factor-time + spatial GP	47	−1260.8	−17.2
Linear-time + spatial GP	27	−1271.4	−27.8
No-time + spatial GP	21	−1305.0	−61.4

✓ Conclusion: AR(1) + spatial GP provides the best predictive fit (highest PSIS-LOO elpd).

Key takeaway

A good spatiotemporal model balances biological realism, predictive performance, and computational simplicity.

bayesAgriFlow Remote-Sensing Bayesian Models Cheatsheet

2/5 — Vegetation Indices and Sensor Calibration

Transforming spectral bands into interpretable agronomic predictors

bayesAgriFlow

① Why indices?

- Reduce raw-band complexity.
- Connect with plant physiology.
- Support phenotyping across time and space.
- Enable interpretable Bayesian modeling.

② Core index function

`bayes_rs_indices()`

Supported indices:

- ✓ **NDVI** — Normalized Difference Vegetation Index
- ✓ **EVI** — Enhanced Vegetation Index
- ✓ **SAVI** — Soil-Adjusted Vegetation Index
- ✓ **NDRE** — Normalized Difference Red-Edge Index

Accepts reflectance bands and returns index values with uncertainty.

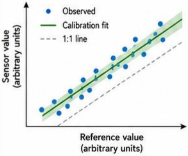
③ Index quick guide

Index	Simple formula	Typical use
NDVI	$\frac{(NIR - Red)}{(NIR + Red)}$	General greenness and vigor
EVI	$2.5 \times \frac{(NIR - Red)}{(NIR + 6 \times Red - 7.5 \times Blue + 1)}$	Dense canopy, reduce saturation & atmosphere
SAVI	$\frac{(NIR - Red)}{(NIR + Red + L)}$ $(L = 0.5)$	Sparse canopy, soil background adjustment
NDRE	$\frac{(NIR - RedEdge)}{(NIR + RedEdge)}$	Chlorophyll sensitivity & nitrogen status

NIR = Near Infrared, Red = Red, RedEdge = Red Edge, Blue = Blue

④ Sensor calibration

`bayes_rs_calibrate()`



- Fit calibration line between sensor and reference measurements.
- Propagate coefficient uncertainty into downstream models.
- Improve comparability across sensors, dates, and locations.

Outputs calibrated reflectance and coefficient posterior distributions.

⑤ Practical interpretation

- **NDVI** — Good general indicator of plant vigor and canopy density.
- **NDRE** — Sensitive to chlorophyll and nitrogen status; less affected by leaf structure.
- **SAVI** — Best when canopy is sparse or soil background is visible.
- **EVI** — Maintains sensitivity in dense canopy; reduces atmospheric and soil effects.

⑥ Key functions

- `bayes_rs_indices()` — Compute vegetation indices with uncertainty.
- `bayes_rs_calibrate()` — Calibrate sensor to reference and propagate uncertainty.
- `bayes_summary()` — Summarize posterior estimates and diagnostics.

All functions return tidy outputs for modeling and reproducibility.

⑦ Agronomic mini-example

Scenario: UAV multispectral flight over soybean field on 2024-07-15.

Example spectral bands (reflectance)

Plot	Blue	Green	Red	RedEdge	NIR
P1	0.08	0.14	0.06	0.41	0.68
P2	0.07	0.13	0.05	0.38	0.66
P3	0.09	0.15	0.07	0.43	0.70
P4	0.08	0.14	0.06	0.40	0.67
...	...	...	...	...	...

Bands are calibrated and indices computed with uncertainty.

Index outputs (posterior mean ± SD)

- NDVI 0.84 ± 0.03
- EVI 0.62 ± 0.04
- SAVI 0.77 ± 0.03
- NDRE 0.66 ± 0.04

Note: Calibrated NDRE showed the strongest association with tissue N status ($R^2 = 0.68$).

Key takeaway

Useful Bayesian remote-sensing models start with agronomically meaningful predictors, not just raw sensor bands.

bayesAgriFlow Remote-Sensing Bayesian Models Cheatsheet

3/5 — Design-Aware Phenotyping and Measurement Error

Integrating experimental design, sensor features, and predictor uncertainty

bayesAgriFlow

1 Why design-aware?

- Field trials have blocks and treatments.
- Plots are not independent in space.
- Sensor features may be noisy.
- Bayesian models can propagate uncertainty.

2 Phenotyping design object

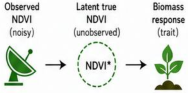
```
bayes_rs_phenotype()
```

Required roles:

- ✓ `trait` → response variable
- ✓ `features` → predictor features (e.g., NDVI, NDRE, height)
- ✓ `x_coord` → x coordinate (easting)
- ✓ `y_coord` → y coordinate (northing)
- ✓ `block / group` → optional grouping (e.g., block, genotype, site)

3 Measurement-error route

```
bayes_rs_model()
```



Known predictor_sd inputs:
Provide per-feature measurement error (e.g., NDVI_sd, NDRE_sd, height_sd).

4 What uncertainty changes

- Wider intervals.
- Less overconfidence.
- Fairer effect estimates.
- More honest prediction.

5 Good model ingredients

- ✓ Informative features.
- ✓ Spatial effect if needed.
- ✓ Temporal term if repeated acquisitions.
- ✓ Clear experimental roles.

6 Key functions

```
bayes_rs_phenotype()
bayes_rs_model()
bayes_fit()
bayes_diagnose()
```

7 Agronomic mini-example

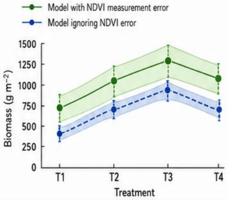
Scenario: Biomass modeled from NDVI, NDRE, and canopy height with block effects.

RCBD layout (4 blocks × 4 treatments)

Block (north → south)

	T1	T2	T3	T4
B1	T3	T1	T4	T2
B2	T2	T4	T1	T3
B3	T4	T2	T3	T1
B4	T1	T3	T2	T4

Biomass response (posterior mean ± 95% interval)



Including NDVI measurement error reduced false certainty while preserving the main signal.

Key takeaway

Remote-sensing models are strongest when sensor features, design structure, and predictor uncertainty are handled together.

bayesAgriFlow Remote-Sensing Bayesian Models Cheatsheet

1/5 — Foundations and Data Workflow

bayesAgriFlow

Analysis-ready UAV, satellite, and proximal-sensor data for Bayesian agronomy

1 What is included?

- Analysis-ready sensor tables with metadata.
- Optional raster extraction to plot or zone level.
- Vegetation indices derived from imagery.
- Map-ready posterior outputs for decision support.

2 Start with data

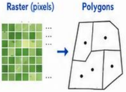
`bayes_rs_data()`

Required roles:

- ✓ `plot_id` → unique plot identifier
- ✓ `x` → x coordinate (easting)
- ✓ `y` → y coordinate (northing)
- ✓ `source` → sensor/source (e.g., UAV, S2)
- ✓ `acquisition date` → date of observation
- ✓ `response or target trait` → observed agronomic response (e.g., biomass, yield)
- ✓ `sensor features` → bands, indices, or features

3 Optional extraction

`bayes_rs_extract()`



Extracted values

plot_id	NDVI
P01	0.72
P02	0.65
P03	0.81
...	...

Uses `terra` *and* `sf` *under the hood. Both are optional dependencies.*

4 Stable workflow

Organize data → **Compute indices** → **Define model** → **Fit** → **Predict / Map** → **Decide**

- **Organize data** — Prepare analysis-ready sensor tables. `bayes_rs_data()`
- **Compute indices** — Derive vegetation and spectral indices. `bayes_rs_indices()`
- **Define model** — Specify response and predictors. `bayes_rs_model()`
- **Fit** — Estimate posterior distributions. `bayes_fit()`
- **Predict / Map** — Generate predictions and uncertainty maps. `bayes_rs_map()`
- **Decide** — Interpret results and take action.

5 Common agronomic uses

- **Biomass prediction** — Estimate biomass or yield potential.
- **Stress screening** — Detect water, nutrient, and heat stress.
- **Disease monitoring** — Identify disease hotspots early.
- **Plot-level phenotyping** — Quantify trait variation across trials.

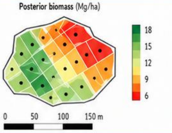
6 Key functions

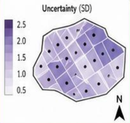
`bayes_rs_data()`	Import and validate analysis-ready remote-sensing data.
`bayes_rs_extract()`	Extract raster values to plots or zones (optional).
`bayes_rs_indices()`	Compute vegetation and spectral indices.
`bayes_rs_model()`	Define model formula and priors for RS variables.
`bayes_rs_map()`	Create posterior prediction and uncertainty maps.

7 Mini-example

Scenario: *UAV-derived NDVI and canopy height for maize biomass.*
Inputs: *plot coordinates, date, NDVI, canopy height.*
Output: *posterior biomass map.*

plot_id	x (m)	y (m)	date	NDVI	canopy_ht (m)	biomass (Mg/ha)
P01	246543	4789123	2024-06-18	0.72	1.48	13.2
P02	246583	4789161	2024-06-18	0.65	1.26	10.6
P03	246612	4789140	2024-06-18	0.81	1.72	15.9
...	...	...	...	...	...	...





Key takeaway

bayesAgriFlow 1.3.0 begins after preprocessing and focuses on scientific analysis of analysis-ready remote-sensing variables.

bayesAgriFlow Spatiotemporal Models Cheatsheet

4/5 — Prediction, Forecasting, and Threshold Probabilities

From posterior estimates to practical field-risk products

1 Prediction vs forecasting

Prediction

Estimate the variable at observed or mapped times and locations.

Forecasting

Predict the variable at future times *(beyond the last observation).*

2 Posterior prediction

```
bayes_st_predict()
```

Outputs:

- Posterior mean (expected value)
- Posterior interval (e.g., 50%, 95%)
- Map-ready grid aligned to space and time

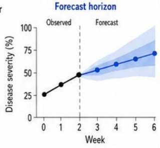
3 Short-term forecasting

```
bayes_st_forecast()
```

Produces forecasts for future times with uncertainty.

Use for:

- Disease risk
- Canopy temperature
- Soil water



4 Threshold probabilities

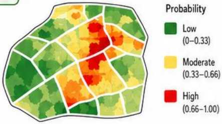
```
bayes_st_exceedance()
```

Compute P(variable > threshold) or P(variable < threshold).

Example thresholds:

- Pr(severity > 20%)
- Pr(soil moisture < 0.12)

Example: Probability disease severity > 20%



5 Communicating uncertainty

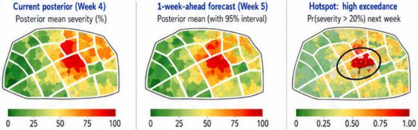
- Use intervals (e.g., 50%, 95%) to convey uncertainty.
- Show posterior standard deviation maps to highlight uncertainty spatially.
- Report threshold probabilities *to support risk-based decisions.*
- Avoid over-interpreting *single-point predictions.*

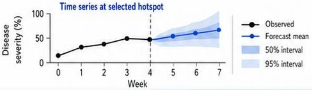
6 Key functions

`bayes_st_predict()`	Posterior prediction at observed/mapped times.
`bayes_st_forecast()`	Short-term forecasts into the future.
`bayes_st_exceedance()`	Threshold exceedance probabilities.
`bayes_predict()`	Non-spatiotemporal prediction.

7 Agronomic mini-example

Scenario: Weekly disease severity.





High probability of exceeding 20% severity next week.

Key takeaway

Prediction tells you what is likely now; forecasting tells you what may happen next.

bayesAgriFlow Remote-Sensing Bayesian Models Cheatsheet

5/5 — Prediction, Uncertainty Mapping, and Change Analysis

Turning Bayesian remote-sensing models into map-ready decisions

bayesAgriFlow

1 Posterior products

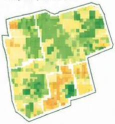
- **Mean predictions**
  Expected value of the response at each location.
- **Interval maps**
  Credible intervals (e.g., 80% or 95%) quantify uncertainty.
- **Threshold probabilities**
  Probability that the response exceeds a user-defined threshold.
- **Change maps**
  Difference between times with uncertainty.

2 Prediction functions

`bayes_rs_predict()`

`bayes_rs_map()`

Example: posterior mean biomass (Mg/ha)
Day 20 prediction



Biomass (Mg/ha)
0 – 2
2 – 4
4 – 6
6 – 8
8 – 10
10 – 12
12 – 14
14 – 16
> 16

3 Uncertainty products

`bayes_rs_uncertainty()`

- **Posterior SD maps**
  Standard deviation of the posterior at each location.
- **Credible interval maps**
  Lower and upper bounds (e.g., 80% or 95%) shown as maps.
- **Threshold probability maps**
  Probability that the response exceeds a user-defined threshold.

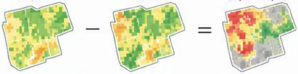
Uncertainty is location-specific and must be reported.

4 Change detection

`bayes_rs_change()`

Compares posterior distributions between two times.

Time 1 (Day 20) − Time 2 (Day 35) = Change (Day 35 – Day 20)



Change legend: Decrease | Stable (± small) | Increase

Summarize change with uncertainty (e.g., probability of increase).

5 How to communicate maps

- **Show legend and units**
  Every map must include units, scale, and a clear legend.
- **Report uncertainty**
  Always accompany predictions with SD, credible intervals, or threshold probabilities.
- **Define thresholds explicitly**
  State the threshold value, units, and rationale.
- **Separate change from raw level**
  Do not interpret change maps as absolute levels.

6 Key functions

`bayes_rs_predict()`

Generate posterior simulations or summaries at observation or grid locations.

`bayes_rs_uncertainty()`

Compute posterior SD, credible intervals, and threshold probability maps.

`bayes_rs_map()`

Create map-ready rasters (mean, quantiles, probabilities) with metadata.

`bayes_rs_change()`

Quantify change between times with uncertainty and generate change maps.

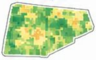
7 Agronomic mini-example

Scenario: Two UAV flights over a bean trial.
Response: Aboveground biomass (Mg/ha).

Posterior mean biomass Day 20

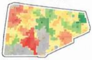
Biomass (Mg/ha)
0 – 2 | 2 – 4 | 4 – 6 | 6 – 8 | 8 – 10 | 10 – 12 | 12 – 14 | 14 – 16 | > 16

Posterior mean biomass Day 35



Biomass (Mg/ha)
0 – 2 | 2 – 4 | 4 – 6 | 6 – 8 | 8 – 10 | 10 – 12 | 12 – 14 | 14 – 16 | > 16

Change map
Day 35 – Day 20



Change (Mg/ha)
Decrease (< −1)
Stable (−1 to 1)
Increase (1 to 3)
Strong increase (> 3)
No data

Red zones: stagnation or loss. *Light green: moderate increase.* *Dark green: rapid increase.*

Key takeaway

Map-ready posterior outputs are valuable only when prediction, uncertainty, and change are interpreted together.

bayesAgriFlow Remote-Sensing Bayesian Models Cheatsheet

4/5 — Multi-Sensor Fusion and Space–Time Integration

Combining UAV, satellite, and ground data in a coherent Bayesian workflow

bayesAgriFlow

1 Why fuse sensors?

- Complementary resolution captures fine and broad patterns.
- Broader coverage across space and time.
- Repeated monitoring improves robustness and early detection.
- Better prediction than single-source models.

2 Fusion design object

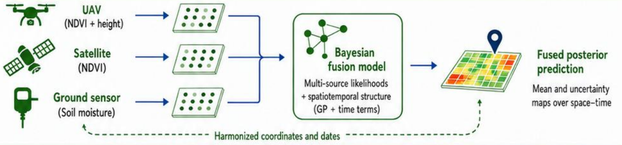
`bayes_rs_fusion()`

Example sources:

UAV	→	NDVI + height
Satellite	→	NDVI
Ground	→	Soil moisture

All sources mapped to **harmonized coordinates and dates.**

3 Concept diagram



4 Where space–time enters

- Acquisitions vary by location and date.
- Spatial GP handles field structure and spatial correlation.
- Temporal terms track dynamics and trends.
- The same spatiotemporal architecture can be reused across variables.

5 Practical cautions

- Avoid duplicated information and overly similar predictors.
- Align units, scales, and preprocessing across all sources.
- Check missingness patterns and coverage gaps.
- Compare fused vs single-source models to verify added value.

6 Key functions

`bayes_rs_fusion()`

`bayes_spatiotemporal()`

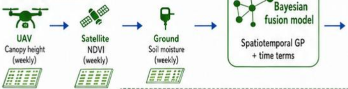
`bayes_compare_models()`

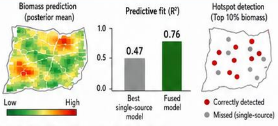
`bayes_predict()`

See series page 1/5 for role definitions.

7 Agronomic mini-example

Scenario: Predict cotton biomass by fusing UAV canopy height, satellite NDVI, and ground moisture.





Fusion improved predictive fit and hotspot detection by combining complementary strengths.

Key takeaway

Sensor fusion is most valuable when each data source contributes unique, auditable information.